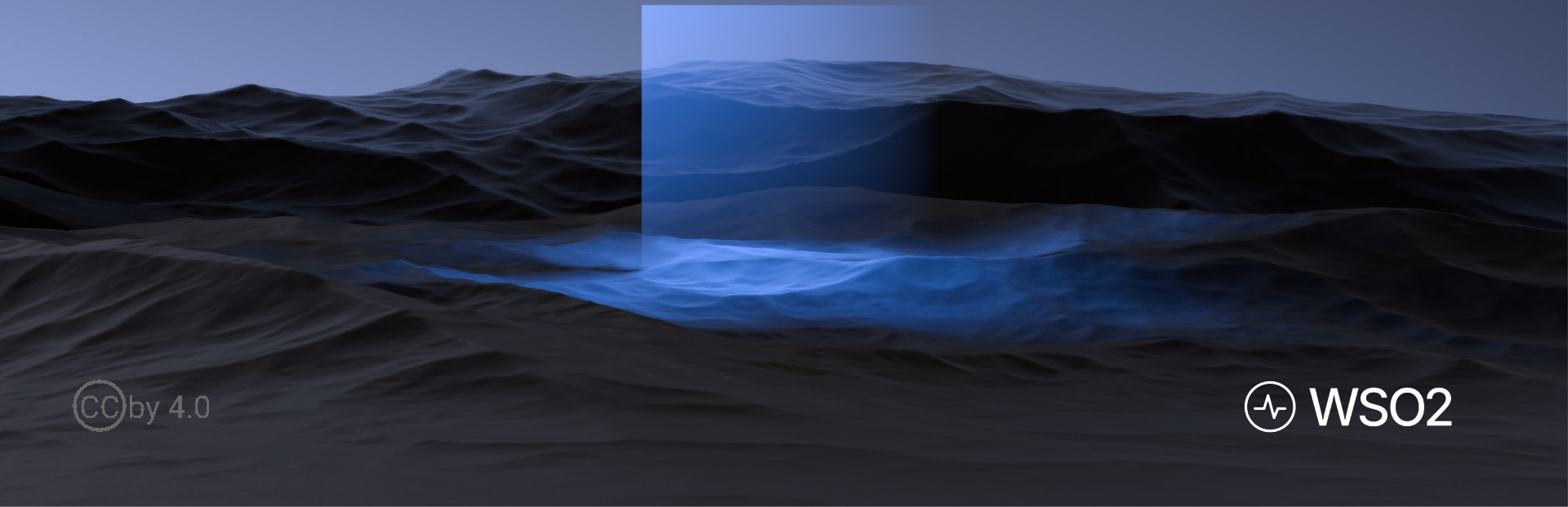
Deployment Patterns



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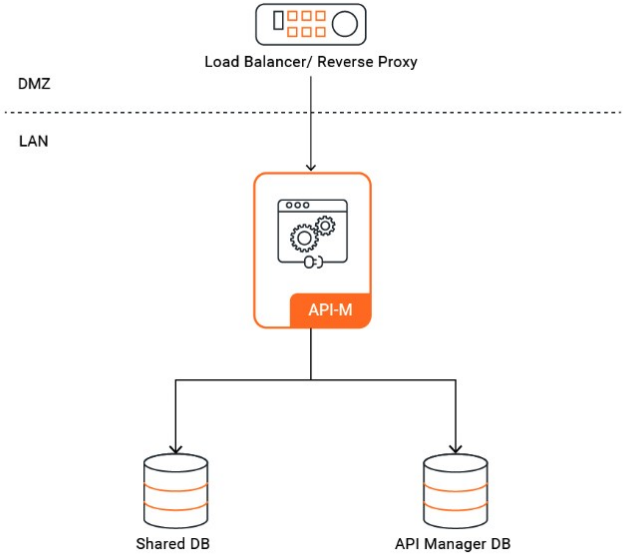
WSO2

Deployment Patterns

WSO2

Pattern #0: All-in-One Setup

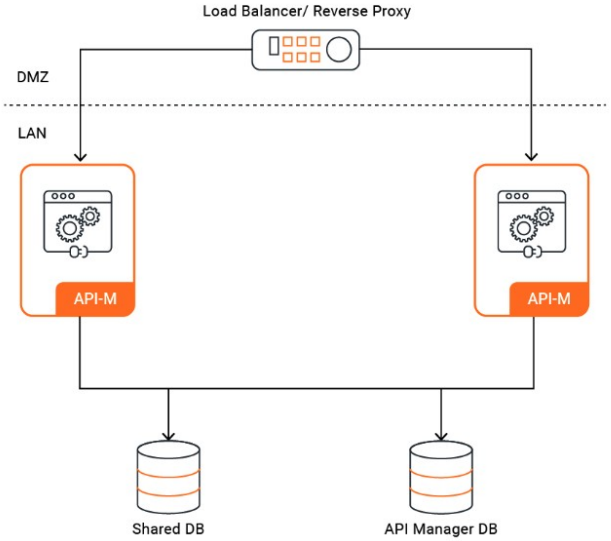
This deployment consists of a single API Manager node handling all functionality. You can use this pattern to development environments or small-scale deployments with low traffic.
This is not recommended for production use.



-M Deployment with All-in-One Setup

Pattern #1 - All-in-One HA Setup

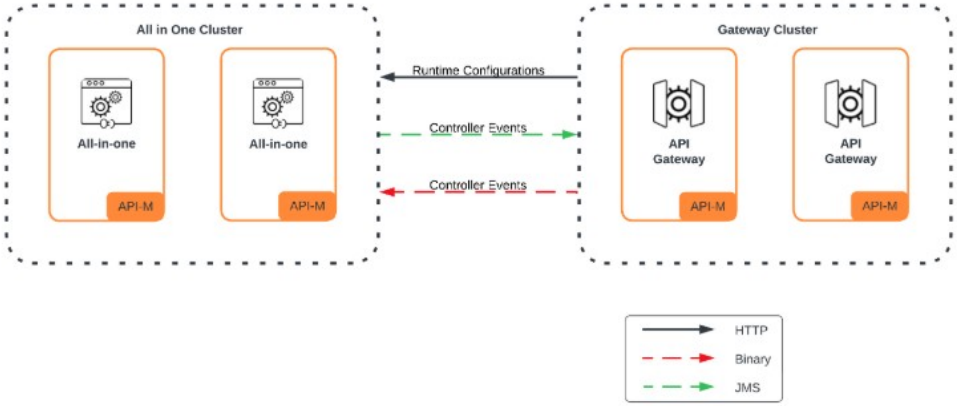
This is a high availability deployment with multiple API Manager nodes in active-active configuration. You can use this pattern for production environments requiring high availability but with moderate traffic



API-M Deployment with All-in-One HA Setup

Pattern #2 - Simple Scalable Deployment

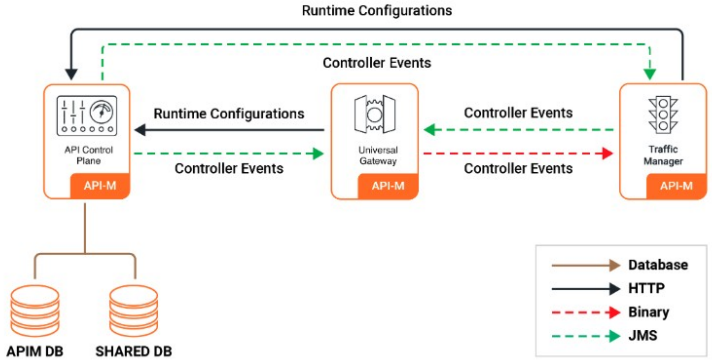
This pattern allows you to deploy with separate gateway nodes and a control plane. However, this setup can easily be scaled. This can be used for environments with higher API traffic needing gateway scalability



Simple Scalable Setup

Pattern #3 - Distributed API-M Deployment with Gateway and Traffic Manager Separated from the Control Plane

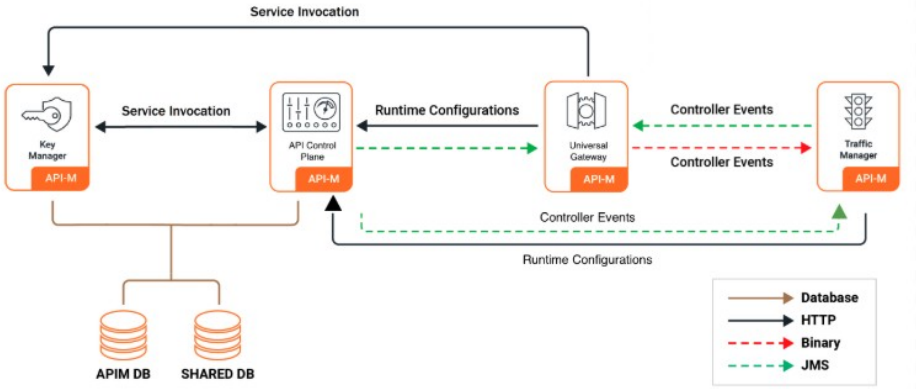
This pattern is for a distributed deployment with separate API Control Plane, Traffic Manager, and Gateway components. This can be used for production environments with high traffic needing component-level scalability.



Distributed API-M Deployment with Gateway and Traffic Manager

Pattern #4 - Fully Distributed Setup

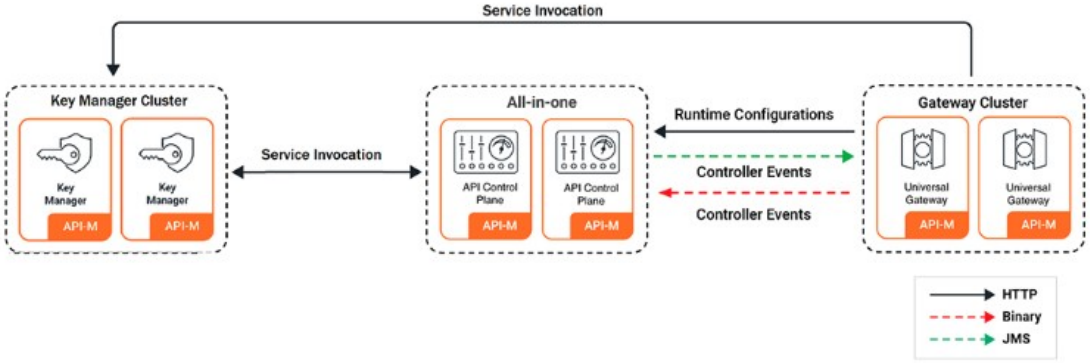
This pattern is for a fully distributed deployment with separate Key Manager component. This can be used for large-scale production environments with complex security requirements



Fully Distributed Setup

Pattern #5 - Simple Scalable Setup with Key Manager Separated

This pattern is for a deployment with separate Gateway and Key Manager components. This can be used for environments focusing on API security with dedicated Key Manager component.

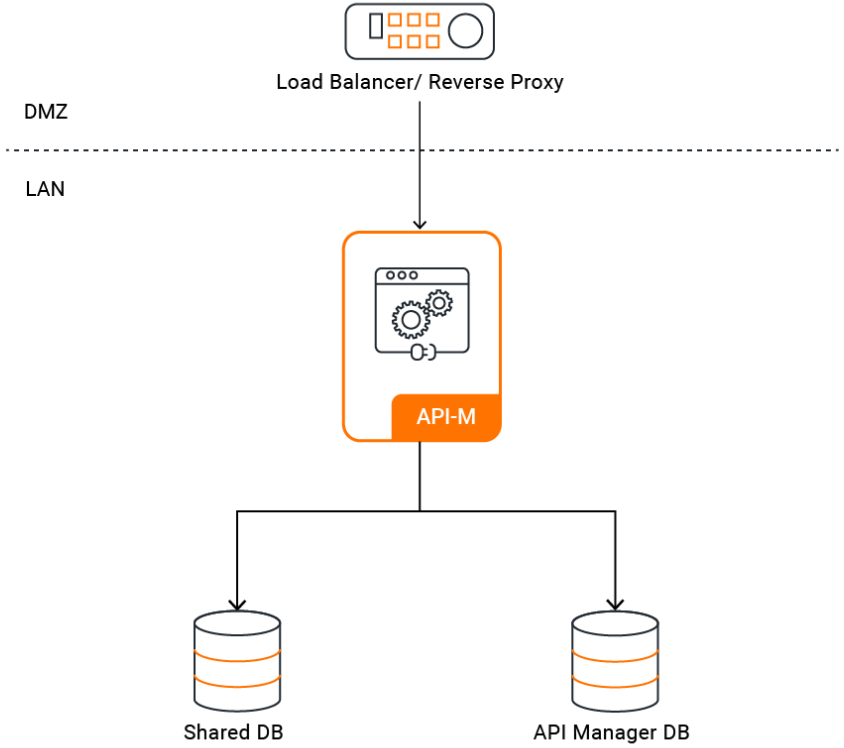


Simple Scalable Setup with Key Manager Separated

Deploying API Manager Using Single Node Instances

WSO2

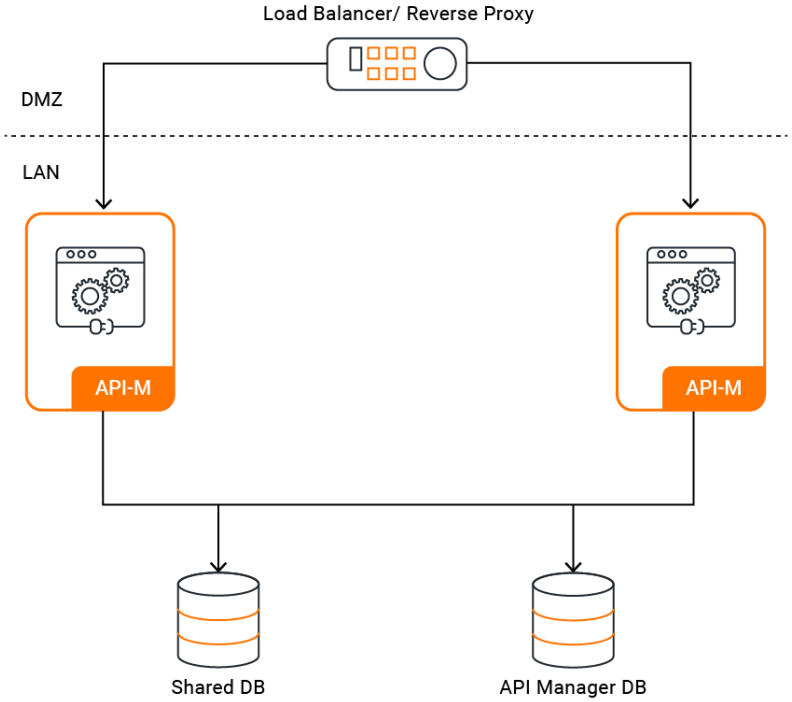
Single Node deployment



API traffic is served by single all-in-one instance of WSO2 API Manager

Configuring a Single Node

Active - Active deployment



API traffic is served by two, all-in-one instances of WSO2 API Manager.

Configuring an Active-Active Deployment

Distributed Deployment

WSO2

Product Component Distributions

- When a WSO2 product starts, it starts all components, features and related artifacts bundled with it.
- But when deploying API Manager in a distributed manner, every node does not need to have all the components.

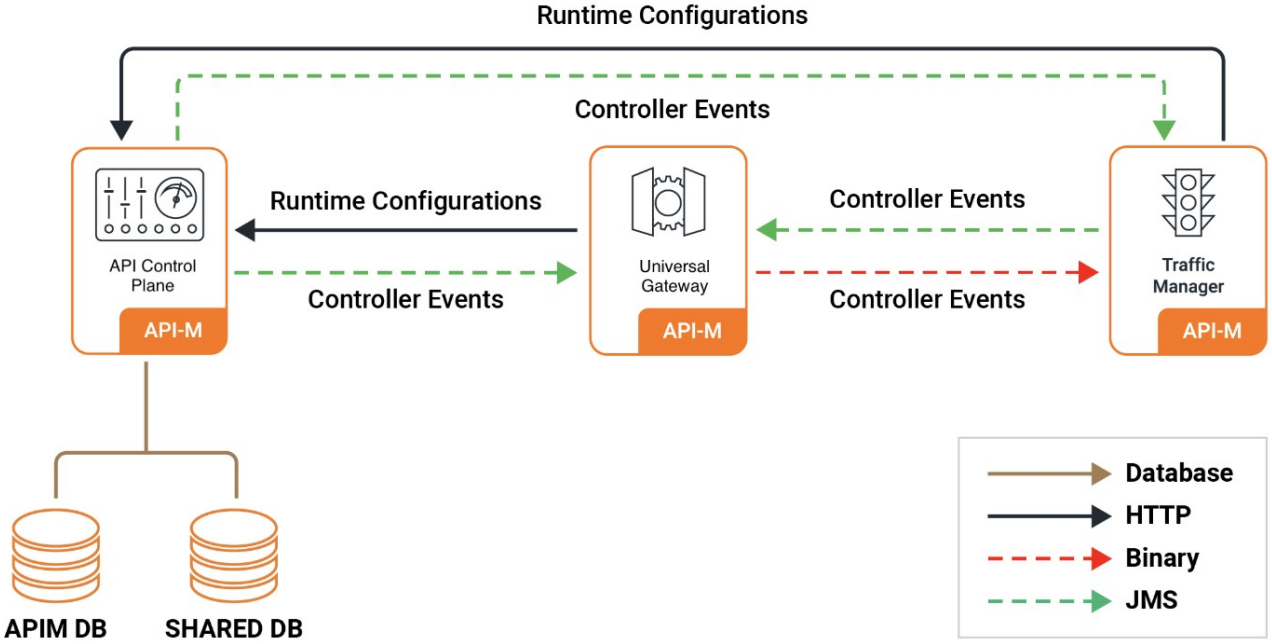
  Eg: Gateway does not need Publisher/ Devportal components etc.
- For this, API Manager offers 3 component distributions as follows.

  WSO2 API Control Plane

  WSO2 Classic (Universal) Gateway

  WSO2 Traffic Manager
- When running, each component will load only the required OSGi bundles and components to that specific node.

Creating the Distributed Deployment



Understanding the Distributed Deployment of WSO2 API-M

Databases

WSO2 API Manager components use the databases as follows:

	API Manager Database
	`apim_db`
	`WSO2AM_DB`
		Shared Database
		`shared_db`
		`WSO2SHARED_DB`

Classic Gateway	Not Used	Not Used
API Control Plane	Used	Used
Traffic Manager	Used	Used

API-M Distributed Deployment Overview

Creating the Distributed Deployment

The three component distributions are set up in a distributed deployment based on the selected pattern

The two databases are connected to the relevant components respectively

Notice the port offset and the HTTPS port of each WSO2 server component*

The default communication protocol of Key Manager is WSClient

The entire setup will also be fronted by a Load Balancer

*If multiple nodes deploy in the same physical machine or vm.

Deploying API-M in a Distributed Setup

Let’s try it out!

Deploy WSO2 API Manager Pattern #1

WSO2